Experiment-11

Write a python program for Map Coloring to implement CSP

Aim

To write and execute a Python program to colour the regions of a map using a Constraint Satisfaction Problem (CSP) with backtracking, so that no two neighbouring regions have the same colour.

Algorithm

1. Take the regions WA, NT, SA, Q, NSW, V and T as the variables.
2. Take the colours Red, Green and Blue as the domain of each variable.
3. Define the constraint that two adjacent regions must not have the same colour.
4. Select the next region that has not been coloured.
5. Try each colour and check whether it is safe with respect to its neighbours.
6. If the colour is safe, assign it and move to the next region.
7. If no colour is possible, backtrack by removing the last assignment and try another colour.
8. Continue until all the regions are coloured.
9. Display the colour assigned to each region.

Coding

ex-11.py - D:/ex-11.py (3.10.0)

File Edit Format Run Options Window Help

```
# Map Coloring Problem using CSP (Backtracking)

regions = ["WA", "NT", "SA", "Q", "NSW", "V", "T"]

neighbours = {
    "WA": ["NT", "SA"],
    "NT": ["WA", "SA", "Q"],
    "SA": ["WA", "NT", "Q", "NSW", "V"],
    "Q": ["NT", "SA", "NSW"],
    "NSW": ["Q", "SA", "V"],
    "V": ["SA", "NSW"],
    "T": []
}

colours = ["Red", "Green", "Blue"]

# A colour is safe if no neighbour already has the same colour
def is_safe(region, colour, assignment):
    for n in neighbours[region]:
        if assignment.get(n) == colour:
            return False
    return True

def solve(assignment):
    if len(assignment) == len(regions):
        return assignment

    region = regions[len(assignment)]
    for colour in colours:
        if is_safe(region, colour, assignment):
            assignment[region] = colour
            result = solve(assignment)
            if result:
                return result
            del assignment[region]       # backtrack
    return None
```

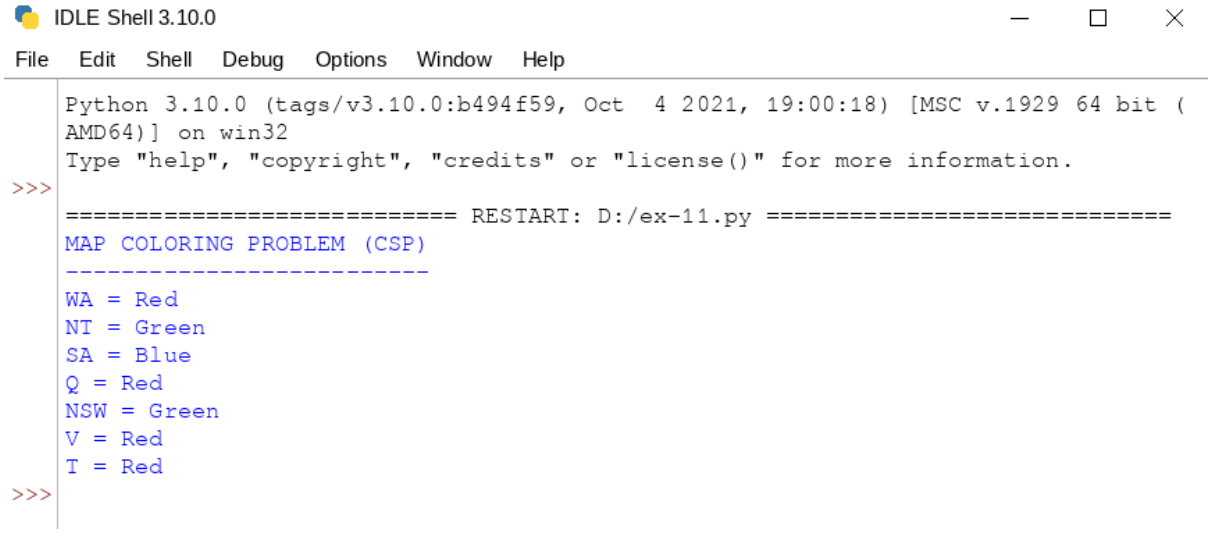
ex-11.py - D:/ex-11.py (3.10.0)

File Edit Format Run Options Window Help

```
print("MAP COLORING PROBLEM (CSP)")
print("--------------------------")
solution = solve({})

if solution:
    for region in regions:
        print(region, "=", solution[region])
else:
    print("No solution exists.")
```

Output



Result

The Map Coloring problem was successfully solved using CSP with backtracking. All the regions were coloured using only three colours without any two neighbouring regions having the same colour.